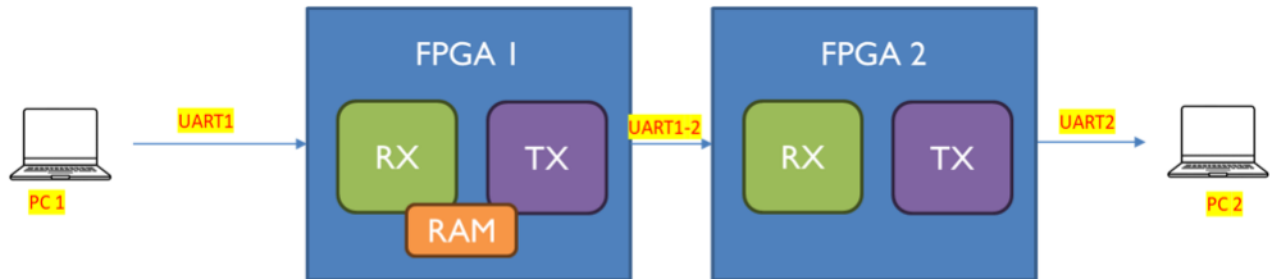


Assignment 3: Interfaces and memories (Group work)

The objective of this exercise is to design and implement two digital circuits using **two separate FPGAs**. Each FPGA shall be equipped with a **UART interface** to enable communication with a host PC for transmitting and receiving messages.

The overall system architecture is illustrated in the diagram below and enables end-to-end data communication between two PCs through the FPGAs.



Data Flow

The complete data flow of the system is as follows:

PC1 → UART1 → FPGA 1 (RX) → RAM → FPGA 1 (TX) → UART1-2 → FPGA 2 (RX) → FPGA 2 (TX) → UART2 → PC2

In this setup:

- FPGA 1 receives data from PC1 via a UART receiver (RX), data is the group number (e.g., GROUP 1)
- Processes and forwards the data internally,
- Then retransmits it using a UART transmitter (TX),

Bonus Task

As a bonus extension, replace the **UART1-2 communication link** between the two FPGAs with an **SPI protocol**, and demonstrate successful data transfer using SPI instead of UART.

Exercise Report Requirements

Document the outcome of this exercise in a structured technical report that addresses the following points:

1. **Component Description**
Explain the components used in the design and their roles within the system.
2. **Circuit Operation**
Describe how the circuit works, including the main block design and its interactions.
3. **Data Flow Analysis**
Present the complete data flow from **PC1 to PC2**, clearly explaining each step in the process.
4. **Simulation Results**
Perform and document the following three types of simulations:
 - Functional simulation
 - Post-synthesis simulation
 - Implementation timing simulation

Comment on and compare the results of the three simulations.

Support your explanations by including **relevant code excerpts**, **screenshots from the simulator**, and **photos/screenshots from the FPGA hardware** where appropriate.

Submission Guidelines

- **One submission per group**
- The report must include:
 - Group number
 - Your name(s) and student ID(s)
- **Maximum length:** 3 pages
- **Format:** PDF
- **Additional requirement:**
Attach a video demonstration or provide a shared video link (e.g., YouTube).